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Automatic Classification of Driving Maneuvers via Supervised Learning

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***Abstract*—This study presents a supervised learning approach for the automatic classification of driving maneuvers using only vehicular dynamics data collected through a Racelogic VBOX 3i logger at 100 Hz. The methodology involved (1) spatial segmentation of a test track via GPS to label five maneuver types—curve with banking, curve without banking, straight line, roundabout, and slalom; (2) training classical (Random Forest, KNN, MLP) and sequential (RNN, LSTM) models; and (3) comparative performance evaluation. Among all tested algorithms, the bidirectional LSTM achieved the highest average F1-score of 95.9%, outperforming the best classical model (MLP) by more than 1.5 percentage points. The classification of roundabout maneuvers reached an F1-score of 95.5%, while slalom was the most challenging with 94.8%. These results demonstrate the effectiveness of temporal modeling in maneuver classification and support its application in intelligent driving assistance, usage-based insurance, and behavioral profiling—without requiring multimodal sensor fusion.**

***Index Terms*—Vehicular maneuver classification, Neural networks, ADAS (Advanced Driver Assistance Systems), Supervised learning, Machine learning, GPS-based segmentation, Driving patterns.**

I. Introduction

Automatic classification of driving maneuvers is essential for the development of Advanced Driver Assistance Systems (ADAS), which can alert the driver to risky situations and improve vehicle stability [1]. By segmenting and labeling trajectory fragments, it is possible to train models that recognize dynamic driving patterns and provide warnings in the presence of hazardous behavior.

Although numerous studies combine diverse data sources (e.g., inertial sensors, onboard video, and CAN-Bus telemetry), in this work, all recordings come exclusively from the VBOX 3i @ 100 Hz Data Logger by Racelogic [2]. The GPS signal recorded by the same device enabled the spatial division of the route into georeferenced segments, allowing for the automatic assignment of maneuver labels [3].

Based on these labels, several supervised learning models—including k-Nearest Neighbors [4], Random Forest [5], and recurrent neural networks (RNN and LSTM) [8]—were trained to evaluate their ability to recognize maneuver types in real time. The accurate classification of these maneuvers is critical for practical applications in fleet monitoring, usage-based insurance, and intelligent driving assistance systems.

This work proposes an experimental pipeline consisting of three main stages: (1) spatial segmentation of the test track for automatic maneuver labeling, (2) training of supervised classification models using exclusively the data captured by the VBOX3i device, and (3) comparative evaluation of model performance through cross-validation techniques and metric-based analysis.

State-of-Art

Driver behavior classification has been extensively addressed through machine learning and deep neural networks, with applications in road safety, usage-based insurance (UBI), and ADAS systems [9], [16]. Bouhsissin et al. [9] emphasize the use of supervised models to identify driving styles, although a lack of consensus remains regarding their definitions and segmentation criteria.

Several approaches have focused on detecting specific maneuvers [15] or risk levels [10], employing driving simulators [10], vehicle sensors (OBD-II, CAN-BUS), and mobile devices [11]. Recent models such as GLMDriveNet [13] and SIFDriveNet [14] integrate visual and vehicular signals using multimodal architectures, achieving high accuracies, though with limited generalization due to closed or artificial environments.

Despite the progress in deep learning (CNNs, LSTMs), classical algorithms like KNN and Random Forest remain useful in resource-constrained scenarios [12], [17]. However, many studies share common challenges such as access to diverse datasets, limited reproducibility, and the difficulty of real-time model deployment [18], [19].

In contrast, the present work proposes a model trained with real-world data segmented by maneuver type, contributing to system generalizability without requiring additional sensors or controlled environments.

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In summary, although previous studies have explored both traditional and deep learning techniques for driving behavior recognition, many rely on simulations, multimodal sensor fusion, or abstract behavioral definitions. This work specifically addresses the classification of distinct driving maneuvers—such as curves, straight segments, roundabouts, and slalom—using data collected under real-world conditions. The following section details the spatial segmentation strategy and the supervised learning models applied to this task.

II. Materials and Methods

This non-experimental, observational, and cross-sectional study aimed to analyze driver behavior in real-world traffic conditions without modifying the environment or introducing experimental stimuli. Participants executed specific driving maneuvers while physiological and telemetry data were collected within a single time frame, allowing the identification of relevant patterns and relationships between variables associated with road safety.

Given the complexity of driving—which integrates perceptual, cognitive, and motor skills—a multivariate analytical approach was adopted. This enabled the simultaneous analysis of several variables, the identification of group-level differences among drivers, and the exploration of latent patterns not easily captured using traditional methods. Sociodemographic data such as age, gender, educational level, driving experience, and area of residence were collected to support the analysis of potential correlations with dynamic driving behavior and to identify high-risk driver profiles.

The study included individuals over 18 years old with good visual and motor health, a valid driver's license with at least one year of experience, and who voluntarily signed an informed consent form.

A. Equipment and Data Acquisition

The tests were conducted using a 2018 Kia Rio, EX version. For driving data collection, an integrated acquisition system was employed based on the VBOX 3i — 100Hz Data Logger from Racelogic [2]. This device recorded both vehicle dynamics data and information from theCAN-Bus system. It also captured geolocation data using GPS antennas installed on the test vehicle. The data were collected at a sampling frequency of 100 Hz.

B. Classification Algorithms

Five supervised models were implemented: Random Forest, K-Nearest Neighbors (KNN), a multilayer perceptron (MLP), a recurrent neural network (RNN), and a bidirectional LSTM network. Their underlying principles and key functionalities are summarized below.

Random Forest

Random Forest is an ensemble method that combines multiple decision trees trained on random subsets of the training data. The final prediction is obtained through majority voting among the trees [5]:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \ldots, T_n(x)\} \tag{1}$$

K-Nearest Neighbors (KNN)

KNN assigns a class to a new sample based on the labels of its k nearest neighbors in the feature space [4], with Euclidean distance being the most commonly used metric:

$$d(x, x_i) = \sqrt{\sum_{j=1}^{n} (x_j - x_{ij})^2}, \quad \hat{y} = \text{mode}\{y_{(1)}, \ldots, y_{(k)}\} \tag{2}$$

Multilayer Perceptron (MLP)

The multilayer perceptron (MLP) is a feedforward neural network comprising an input layer, one or more hidden layers, and an output layer. Each neuron applies a linear transformation followed by a nonlinear activation function, typically ReLU. The operation of layer l is defined as:

$$h^{(l)} = f(W^{(l)} h^{(l-1)} + b^{(l)}) \tag{3}$$

where $W^{(l)}$ and $b^{(l)}$ are the weights and biases of the layer, and f is the activation function. The model is trained using backpropagation [6] and the Adam optimizer. It is suitable for classification tasks involving fixed-length data windows, and its simplicity makes it appropriate for resource-constrained systems, such as embedded platforms.

Recurrent Neural Network (RNN)

Recurrent Neural Networks (RNNs) model temporal sequences by maintaining information from previous time steps [7]. Their basic recurrence relation is:

$$h_t = \tanh(W_{ih} x_t + W_{hh} h_{t-1} + b) \tag{4}$$

where h_t encodes the system state up to time t.

Long Short-Term Memory (LSTM)

LSTM networks improve upon traditional RNNs by incorporating gates that regulate the flow of information [8]. Their behavior is defined by:

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f) & i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i) \\ o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o) & \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t & h_t &= o_t \odot \tanh(c_t) \end{aligned} \tag{5}$$

III. Classification and Visualization of Driving Maneuvers

The classification of driving maneuvers was carried out through spatial segmentation of the track, based on the identification of circuit zones where different types of maneuvers were performed. These sections were delineated using GPS coordinates, based on the analysis of the trajectory and the characteristic shape of each maneuver. The identified maneuvers were as follows:

- **Straight**: Sections of the track with linear trajectories and no significant curvature.
- **Banked Turn**: Segments with curvature and road inclination that allow the vehicle to take the turn at higher speed and with more stability.

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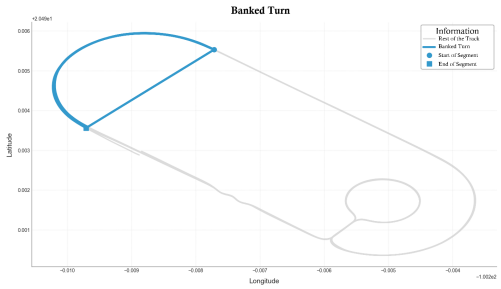
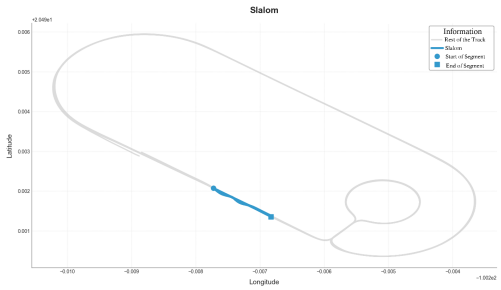
- **Flat Turn**: Sections with sustained curvature but no structural inclination in the pavement.
- **Slalom**: A maneuver in which the vehicle must weave around a series of aligned cones, making alternating direction changes in short distances. This tests the driver's agility and the vehicle's ability to respond to rapid trajectory changes.
- **Roundabout**: Circular section of the track where the vehicle performs a prolonged turn. **Note:** The red section in Fig. 2 was excluded because it corresponds to the approach to the roundabout, introducing non-representative data that could affect the classifier.

A. *Visualization of Individual Maneuvers*

The following figures illustrate examples of how the track sections were segmented by maneuver type. In Fig. 1, the labeled segments for the banked turn (a) and slalom (b) maneuvers are shown.



(a) Example visualization of the **Banked Turn** maneuver. The corresponding segment is highlighted in blue over the full track layout.



(b) Example visualization of the **Slalom** maneuver. The highlighted section shows alternating direction changes between left and right in a controlled space.

Fig. 1: Spatial segmentation of maneuvers based on GPS coordinates on the test track.

Figure 2 shows an overview of the route with all identified and labeled maneuvers, allowing the spatial distribution of each type of maneuver across the track to be visualized.

After classifying the maneuvers using the procedure described above, the resulting distribution of records per maneuver type was obtained, as shown in Table I. This distribution reveals a class imbalance.

Due to the class imbalance, stratified sampling was applied during training to ensure fair representation of each maneuver. Although techniques such as SMOTE (*Synthetic Minority Over-sampling Technique*) can generate synthetic examples,

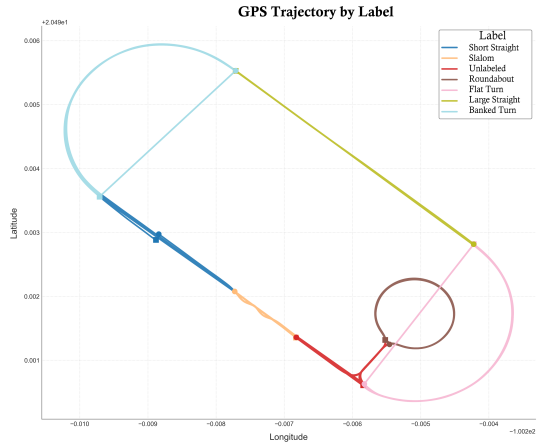


Fig. 2: Visualization of all labeled maneuvers along the route. Each maneuver segment is represented in a different color according to its category.

TABLE I: Distribution of Records by Maneuver Type

Maneuver	Records
Flat Turn	261,732
Banked Turn	245,237
Straight	328,003
Slalom	137,048
Roundabout	318,523
Excluded Data	305,281

they were excluded in this study to avoid introducing bias or noise.

IV. Variables and Model Training

A. *Variable Selection*

For the classification of driving maneuvers, 15 variables were selected from the CAN-BUS system and vehicle dynamics analysis. These include: steering angle, accelerations (longitudinal, lateral, and vertical), angular velocity (yaw, pitch, and roll), RPM, brake and accelerator pedal position, active gear, and individual wheel speeds. All variables were normalized using a *StandardScaler* before being used in the supervised learning models.

B. *Traditional Model Architectures and Training*

Three algorithms were implemented: Random Forest, K-Nearest Neighbors (KNN), and a Multi-Layer Perceptron (MLP) neural network. Model validation was carried out using two methods: Hold-Out (80-20), with 80% of the dataset used for training and 20% for testing, and 5-fold Cross-Validation (KFCV) with $K = 5$.

The configurations used for each model were as follows:

- **Random Forest**:
  - 100 trees, with a maximum depth of 10.
  - 150 trees, with a maximum depth of 10.
- **K-Nearest Neighbors (KNN)**:
  - $k = 3$, 5, and 7 values.

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- **Multi-Layer Perceptron Neural Network (MLP)**:
  - 1 hidden layer with 50 neurons.
  - 2 hidden layers: 100 neurons in the first layer, 50 in the second.

C. Recurrent Neural Network-Based Models (RNN and LSTM)

For sequential models used in maneuver classification, a Recurrent Neural Network (RNN) and a Long Short-Term Memory (LSTM) network were trained on temporal sequences of 50 and 100 samples, preserving the sequential structure of vehicle data.

1) RNN Model

To complement manual maneuver segmentation, a classic Recurrent Neural Network (RNN) model was implemented using PyTorch. This model was trained with the following parameters:

- **Sequence length:** 50 and 100 time steps (two configurations).
- **Layers:** 1 recurrent layer (RNN).
- **Hidden units:** 64 neurons.
- **Batch size:** 64.
- **Number of epochs:** 30.
- **Optimizer:** Adam with a learning rate of 0.001.
- **Loss function:** CrossEntropyLoss.

2) Bidirectional LSTM Model

A bidirectional LSTM architecture was also implemented, suitable for processing longer temporal sequences:

- **Sequence length:** 50 and 100 time steps.
- **Layers:** 2 LSTM layers with 128 units each.
- **Dropout:** 0.3.
- **Batch size:** 128.
- **Number of epochs:** 50.
- **Optimizer:** Adam with a learning rate of 0.001.
- **Loss function:** CrossEntropyLoss.

V. Results

This section presents the best results obtained for the different algorithms using two validation methods: Hold-out 80-20 and K-Fold Cross Validation (KFCV) with $K = 5$, applied to driving maneuver classification. Based on the confusion matrix, the classification metrics *precision*, *recall*, and *F1-score* were calculated.

A. Classical Models

a) Random Forest

Two configurations were evaluated: 100 and 150 trees, using both hold-out and K-Fold Cross Validation (KFCV). In both cases, results were similar; however, the best performance was achieved with 150 trees, a maximum depth of 10, and KFCV, as shown in Fig. 3. The detailed metrics of precision, recall, and F1-score for each maneuver class are presented in Table II, with an average F1-score of 92.0%.

b) K-Nearest Neighbors (KNN)

The KNN algorithm was tested with three values of k: 3, 5, and 7, using the aforementioned validation methods. Fig. 4 shows the confusion matrix for $k = 3$ with K-Fold validation, which yielded the best performance. Precision, recall, and F1-score metrics for each class are presented in Table III, revealing an average F1-score of 91.4%.

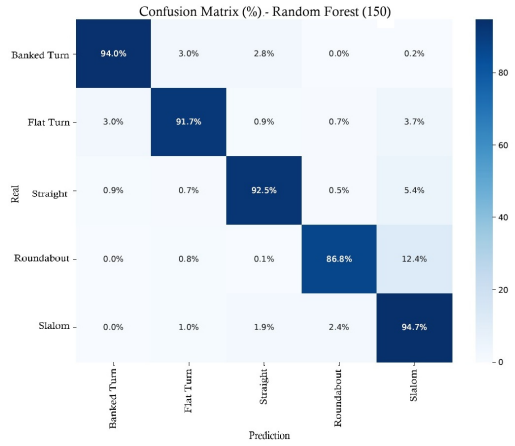

Fig. 3: Confusion matrix for Random Forest with 150 trees.

TABLE II: Evaluation metrics for the Random Forest model (150 trees)

Class	Precision (%)	Recall (%)	F1-Score (%)
Banked_Curve	96.1	94.0	95.0
Unbanked_Curve	94.4	91.7	93.0
Straight	94.2	92.5	93.3
Roundabout	96.0	86.8	91.2
Slalom	81.3	94.7	87.5
Average	**92.4**	**91.9**	**92.0**

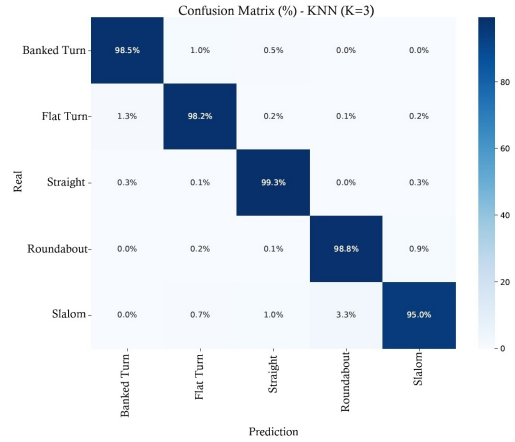

Fig. 4: Confusion matrix for KNN with $k = 3$.

TABLE III: Evaluation metrics for the KNN model ($k = 3$)

Class	Precision (%)	Recall (%)	F1-Score (%)
Banked_Curve	96.6	93.8	95.2
Unbanked_Curve	92.5	92.8	92.6
Straight	90.5	95.0	92.7
Roundabout	88.0	93.5	90.7
Slalom	89.1	82.3	85.6
Average	**91.3**	**91.5**	**91.4**

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c) Multi-Layer Perceptron (MLP)

Both neural network configurations were tested, yielding good results in both. However, better performance was achieved with two hidden layers: 100 neurons in the first layer and 50 in the second. The best performance was obtained with a configuration of two hidden layers (100 and 50 neurons, respectively) and K-Fold cross-validation. The corresponding confusion matrix is shown in Fig. 5. Detailed metrics for each class are presented in Table IV, with an average F1-score of 94.3%.

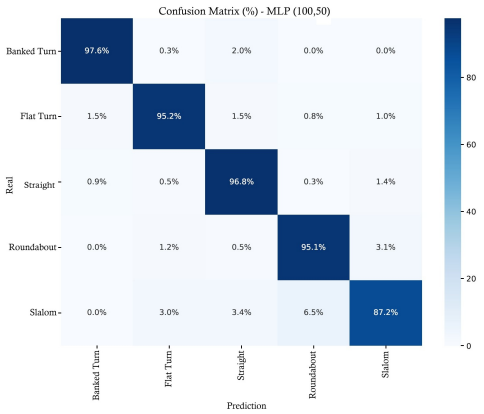

Fig. 5: Confusion matrix for the MLP neural network.

TABLE IV: Evaluation metrics for the MLP model (100,50)

Class	Precision (%)	Recall (%)	F1-Score (%)
Banked_Curve	97.6	97.6	97.6
Unbanked_Curve	95.1	95.2	95.1
Straight	92.1	96.8	94.4
Roundabout	92.8	95.1	93.9
Slalom	93.9	87.2	90.4
Average	**94.3**	**94.4**	**94.3**

B. Sequential Models (RNN and LSTM)

a) RNN with 100-Sample Sequence

Two sequence lengths (50 and 100 samples) were used for the RNN model. Better performance was observed with the 100-sample sequence, as shown in Fig. 6. Evaluation metrics for this configuration are presented in Table V, with an average F1-score of 93.9%.

TABLE V: Evaluation metrics for the RNN model

Class	Precision (%)	Recall (%)	F1-Score (%)
Banked_Curve	98.5	98.8	98.6
Unbanked_Curve	94.2	96.4	95.3
Straight	95.6	95.3	95.4
Roundabout	89.5	93.7	91.5
Slalom	91.9	85.6	88.6
Average	**93.9**	**94.0**	**93.9**

b) Bidirectional LSTM with 100-Sample Sequence

The bidirectional LSTM model also used both 50 and 100-sample sequences. Again, better performance was achieved with the 100-sample sequence, as shown in Fig. 7. The corresponding metrics are detailed in Table VI, achieving an average F1-score of 95.9%, the highest among all evaluated models.

Fig. 6: Confusion matrix for the RNN model.

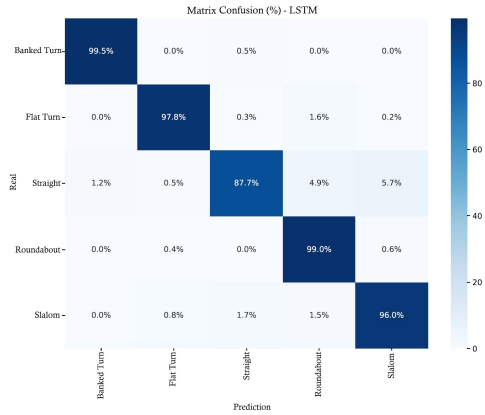

Fig. 7: Confusion matrix for the bidirectional LSTM model.

TABLE VI: Evaluation metrics for the LSTM model

Class	Precision (%)	Recall (%)	F1-Score (%)
Banked_Curve	99.0	99.5	99.2
Unbanked_Curve	98.3	97.8	98.0
Straight	96.7	87.7	92.0
Roundabout	92.3	99.0	95.5
Slalom	93.7	96.0	94.8
Average	**96.0**	**96.0**	**95.9**

C. Comparative Performance

Table VII summarizes the average performance of the classification algorithms used. The results compare the average *Precision*, *Recall*, and *F1-Score* values, highlighting the superior performance of the LSTM network, which can be attributed to its ability to model temporal dependencies.

The results reveal that while classical models such as Random Forest and KNN achieved reasonable performance (F1-scores between 91–92%), they consistently underperformed in the classification of temporally complex maneuvers, especially slalom. This limitation can be attributed to the loss of temporal structure in these models. Since they operate on static data windows, they are unable to capture the sequential nature

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TABLE VII: Average performance of models

Model	Precision	Recall	F1-Score
RF (150)	92.4	91.9	92.0
KNN (k=3)	91.3	91.5	91.4
MLP (100,50)	94.3	94.4	94.3
RNN (100)	93.9	94.0	93.9
LSTM (100)	**96.0**	**96.0**	**95.9**

of vehicle dynamics essential to identify patterns of alternating direction present in slalom maneuvers. Additionally, high intra-class variability and class imbalance—slalom had the fewest samples compared to straight or curve without banking—contributed to lower recall and precision in these classes.

In contrast, the bidirectional LSTM model achieved the best performance across all maneuvers (average F1-score of 95.9%), including the most challenging class. This can be explained by its capacity to learn long-term dependencies and preserve sequence context. The use of 100-sample time windows gave the model sufficient temporal resolution to capture the complete dynamics of each maneuver. Furthermore, the bidirectional architecture allowed the model to incorporate both past and future states, which improved classification of symmetric and complex patterns like slalom. These findings support the superiority of sequential deep learning approaches in tasks involving temporal dynamics such as vehicle maneuver classification.

VI. Conclusions

This study presents a comparative evaluation of classical and sequential machine learning models for the automatic classification of real-world vehicular maneuvers based solely on data acquired from a high-frequency VBOX data logger. The classification pipeline leveraged spatial GPS-based segmentation and focused on five maneuver types: straight, curves with and without banking, roundabout, and slalom.

Among the models evaluated, traditional classifiers such as Random Forest, KNN, and MLP demonstrated competitive performance, achieving average F1-scores between 91.4% and 94.3%. However, recurrent neural networks—particularly the bidirectional LSTM—consistently outperformed classical models, reaching an F1-score of 95.9%. This superior performance is attributed to their ability to model temporal dependencies inherent to driving maneuvers, especially in dynamic scenarios such as slalom and roundabouts.

The findings validate the effectiveness of sequence-aware models for vehicular behavior recognition tasks and support their adoption in intelligent transportation applications, including driver assistance systems, fleet monitoring, and usage-based insurance. The exclusive use of CAN and GPS-derived data also demonstrates the feasibility of implementing maneuver classification systems without additional sensors or complex instrumentation. Future work will address class imbalance, explore real-time deployment, and expand the maneuver taxonomy under diverse traffic and environmental conditions.

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